Nathan Tsao

nathantsao.com | | |

Machine learning research engineer with two years of experience developing reinforcement learning algorithms, generative models, and compositional machine learning frameworks for autonomous systems and embedded devices. Two first-author preprints under review at NeurIPS and CDC.

Education

University of Texas at Austin — Aug 2023 – May 2025
Masters of Science: Mechanical Engineering — GPA: 3.74
Thesis: Neural Port-Hamiltonian Differential Algebraic Equations

University of Illinois Urbana Champaign — Aug 2019 – May 2022
Bachelors of Science: Mechanical Engineering — GPA: 3.87

Work Experience

NASA Ames Research Center — June 2025 – Aug 2025
Autonomous Aircraft Operations Research Intern — Mountain View, CA

- Building a decentralized path-planning framework to encourage traffic following behavior in dense multi-agent air-spaces to improve overall safety and performance.
- Developing GPU-accelerated multi-modal traffic simulation of the Bay Area to guide infrastructure development of unmanned aerial vehicles.

Berkeley Lights — June 2022 – Sep 2022
Hardware Engineering Intern — Berkeley, CA

- Automated temperature calibration and data-acquisition pipelines in Python.
- Developed mechanical and electrical design of temperature calibration sensor of OptoSelect chip.

Research Experience

Autonomous Systems Group: UT Austin — Dec 2023 – May 2025
Graduate Research Assistant — Austin, TX

- Innovated a compositional machine learning modeling framework for differential-algebraic systems, enabling scalable modeling of electrical networks.
- Devised a low-power, low-latency human activity recognition algorithm optimized for batteryless energy-harvesting sensor platforms, resulting in 15-50% relative improvement over baselines.

Hybrid Robotics Group: UC Berkeley — May 2022 – Jan 2023
Visiting Research Assistant — Berkeley, CA

- Implemented a proof-of-concept reinforcement learning-based locomotion balancing controller for tailed quadruped robots.
- Designed and built hardware for a 3D-printed tail that attaches to a quadruped robot, including custom quasi-direct-drive actuators and FOC motor controller.

RoboDesign Lab: UIUC — Jan 2022 – May 2022
Undergraduate Research Assistant — Urbana, IL

- Prototyped a low-cost force-sensing humanoid robot foot using elastomers and Hall sensors.
- Explored the use of Gaussian processes to estimate force from elastomer deformation.

Preprints

Cyrus Neary*, Nathan Tsao*, and Ufuk Topcu. Neural Port-Hamiltonian Differential Algebraic Equations for Compositional Learning of Electrical Networks. *Under review at CDC 2025.*

Geffen Cooper*, Nathan Tsao*, Filippos Fotiadis, Ufuk Topcu, Radu Marculescu. Learning from Sparse and Asynchronous Data Streams for Batteryless Sensors. *Under review at NeurIPS 2025.*

Selected Projects

Causal Diffusion Guidance May 2025

Statistical Machine Learning Final Project

- Developed a diffusion-based framework for generating counterfactual samples from structural causal models.
- Preliminary experiments on causal image datasets demonstrate the ability to generate high-quality, causally consistent counterfactual samples, suggesting the emergence of causal reasoning in diffusion models.

Teaching Experience

ASE 370C Feedback Control Systems: UT Austin Jan 2025 – May 2025

Graduate Teaching Assistant Austin, TX

ME 314D Dynamics: UT Austin Sep 2023 – Dec 2023

Graduate Teaching Assistant Austin, TX

Honors and Awards

Dr. J. Parker Lamb Endowed Presidential Fellowship: UT Austin 2023

Nominated by the Walker Department of Mechanical Engineering.

Highest Honors: UIUC 2022

Awarded to students with at least a 3.8 Illinois GPA and continual commitment to service and education.

Dean's List: UIUC 2019-22

Awarded to undergraduate students in the top 20 percent of their college class.

Outreach

Del Valle High School Visit Mar 2025

Presented machine learning academic research to engineering and robotics high school students. Austin, TX

UT Austin Girls Day Feb 2024

Guided K-8th graders to program remote-controlled cars in Python. Austin, TX

Skills

Skills: Deep Learning | Generative AI | Linux
Programming Languages: Python | Dart | SQL
Frameworks: PyTorch | Jax | Flax | Pandas | Docker | Stable-Baselines3 | Git | Flutter | Supabase
Languages: English | Mandarin
Relevant Coursework: Reinforcement Learning | Statistical Machine Learning | Theoretical Statistics | Convex Optimization